

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.r

```
import torch
torch.cuda.is_available()
```

True

```
import os

# 假设你的项目文件夹在云端硬盘根目录下叫 mini-vm
# 默认路径结构是: /content/drive/MyDrive/你的文件夹名
path = "/content/drive/MyDrive/mini-vm"
```

```
# 切换到项目目录
%cd {path}
```

```
# 查看当前目录下的文件, 确认是否正确
!ls
```

```
/content/drive/MyDrive/mini-vm
mini-vm
```

```
!git clone https://github.com/zxuuustupid/mini-vm.git
```

fatal: destination path 'mini-vm' already exists and is not an empty directory.

```
# 进入项目文件夹
%cd mini-vm/
```

```
# 再进入 dataset 文件夹
%cd dataset/
```

```
# 现在查看内容, 就能看到 dataset 里的文件了
!ls
```

```
/content/drive/MyDrive/mini-vm/mini-vm
/content/drive/MyDrive/mini-vm/mini-vm/dataset
eval_images      pretrain_data.parquet    __pycache__
__init__.py      pretrain_data.parquet.1  sft_data.parquet
lm_dataset.py    pretrain_data.parquet.2  sft_data.parquet.1
```

```
# 下载第一个文件
```

```
!wget https://hf-mirror.com/datasets/jingyaogong/minimind-v_dataset/resolve/main/p
```

```
# 下载第二个文件
```

```
!wget https://hf-mirror.com/datasets/jingyaogong/minimind-v_dataset/resolve/main/s
```

```
# 查看下载结果
```

```
!ls -lh
```

```
--2026-01-03 07:39:41-- https://hf-mirror.com/datasets/jingyaogong/minimind-v_dataset
Resolving hf-mirror.com (hf-mirror.com)... 160.16.86.14
Connecting to hf-mirror.com (hf-mirror.com)|160.16.86.14|:443... connected.
HTTP request sent, awaiting response... 302 Found
```

```
Location: https://cas-bridge.xethub.hf.co/xet-bridge-us/66ffff94008ef654647cefd0/c13f
```

```
--2026-01-03 07:39:42-- https://cas-bridge.xethub.hf.co/xet-bridge-us/66ffff94008ef6
Resolving cas-bridge.xethub.hf.co (cas-bridge.xethub.hf.co)... 18.155.68.125, 18.155.
Connecting to cas-bridge.xethub.hf.co (cas-bridge.xethub.hf.co)|18.155.68.125|:443...
HTTP request sent, awaiting response... 200 OK
```

```
Length: 1245818538 (1.2G)
```

```
Saving to: 'pretrain_data.parquet.2'
```

```
pretrain_data.parquet 100%[=====>] 1.16G 59.3MB/s in 20s
```

```
2026-01-03 07:40:02 (58.8 MB/s) - 'pretrain_data.parquet.2' saved [1245818538/1245818538]
```

```
--2026-01-03 07:40:02-- https://hf-mirror.com/datasets/jingyaogong/minimind-v_dataset
Resolving hf-mirror.com (hf-mirror.com)... 160.16.86.14
```

```
Connecting to hf-mirror.com (hf-mirror.com)|160.16.86.14|:443... connected.
```

```
HTTP request sent, awaiting response... 302 Found
```

```
Location: https://cas-bridge.xethub.hf.co/xet-bridge-us/66ffff94008ef654647cefd0/ec2f
```

```
--2026-01-03 07:40:03-- https://cas-bridge.xethub.hf.co/xet-bridge-us/66ffff94008ef6
Resolving cas-bridge.xethub.hf.co (cas-bridge.xethub.hf.co)... 18.155.68.125, 18.155.
```

```
Connecting to cas-bridge.xethub.hf.co (cas-bridge.xethub.hf.co)|18.155.68.125|:443...
```

```
HTTP request sent, awaiting response... 200 OK
```

```
Length: 572421953 (546M)
```

```
Saving to: 'sft_data.parquet.1'
```

```
sft_data.parquet.1 100%[=====>] 545.90M 60.4MB/s in 8.9s
```

```
2026-01-03 07:40:12 (61.0 MB/s) - 'sft_data.parquet.1' saved [572421953/572421953]
```

```
total 3.6G
```

```
drwx----- 2 root root 4.0K Dec 30 13:13 eval_images
```

```
-rw----- 1 root root 0 Dec 30 13:13 __init__.py
```

```
-rw----- 1 root root 3.8K Dec 30 13:13 lm_dataset.py
```

```
-rw----- 1 root root 1.2G Dec 30 13:20 pretrain_data.parquet
```

```
-rw----- 1 root root 199M Jan 3 07:36 pretrain_data.parquet.1
```

```
-rw----- 1 root root 1.2G Jan 3 07:40 pretrain_data.parquet.2
```

```
drwx----- 2 root root 4.0K Dec 30 13:36 __pycache__
```

```
-rw----- 1 root root 546M Dec 30 13:21 sft_data.parquet
```

```
-rw----- 1 root root 546M Jan 3 07:40 sft_data.parquet.1
```

```
%cd ..
```

```
%cd trainer
```

```
/content/drive/MyDrive/mini-vlm/mini-vlm
```

```
/content/drive/MyDrive/mini-vlm/mini-vlm/trainer
```

```
%cd ..
```

```
/content/drive/MyDrive/mini-vlm/mini-vlm
```

```
# 如果项目里有 requirements.txt
```

```
!pip install -r requirements.txt
```

```
# 或者手动安装一些常见的库（根据你的报错来定）
```

```
# !pip install torch torchvision transformers datasets
```

```
!pip cache purge
```

```
Files removed: 352
```

```
!du -h --max-depth=1 / | sort -hr | head -n 10
```

```
du: cannot read directory '/proc/65/task/65/net': Invalid argument
du: cannot read directory '/proc/65/net': Invalid argument
du: cannot access '/proc/6943/task/6943/fd/4': No such file or directory
du: cannot access '/proc/6943/task/6943/fdinfo/4': No such file or directory
du: cannot access '/proc/6943/fd/3': No such file or directory
du: cannot access '/proc/6943/fdinfo/3': No such file or directory
du: cannot access '/proc/7001': No such file or directory
47G    /
31G    /usr
6.9G   /content
6.1G   /root
1.8G   /tools
1.3G   /opt
156M   /var
51M    /datalab
21M    /python-apt
3.4M   /etc
```

```
import torch
import transformers
import peft
```

```
print(f"Torch 版本: {torch.__version__}")
print(f"Transformers 版本: {transformers.__version__}")
print(f"PEFT 版本: {peft.__version__}")
```

```
Torch 版本: 2.6.0+cu124
Transformers 版本: 4.57.1
PEFT 版本: 0.7.1
```

```
%cd trainer
```

```
/content/drive/MyDrive/mini-vlm/mini-vlm/trainer
```

```
%cd ..
!mkdir out
```

```
/content/drive/MyDrive/mini-vlm/mini-vlm
mkdir: cannot create directory 'out': File exists
```

```
%cd out/
```

```
wget https://modelscope.cn/models/gongjy/MiniMind2-V-PyTorch/resolve/master/llm_512.pth --no-check-certificate
```

```
--2026-01-03 07:49:54-- https://modelscope.cn/models/gongjy/MiniMind2-V-PyTorch/resolve/master/llm_512.pth  
Resolving modelscope.cn (modelscope.cn)... 47.251.62.57  
Connecting to modelscope.cn (modelscope.cn)|47.251.62.57|:443... connected.  
HTTP request sent, awaiting response... 302 Found  
Location: https://cdn-lfs-cn-1.modelscope.cn/prod/lfs-objects/11/39/171ld0fbac13c503f8e58238592/llm_512.pth [application/octet-stream]  
--2026-01-03 07:49:56-- https://cdn-lfs-cn-1.modelscope.cn/prod/lfs-objects/11/39/171ld0fbac13c503f8e58238592/llm_512.pth  
Resolving cdn-lfs-cn-1.modelscope.cn (cdn-lfs-cn-1.modelscope.cn)... 163.181.82.171,  
Connecting to cdn-lfs-cn-1.modelscope.cn (cdn-lfs-cn-1.modelscope.cn)|163.181.82.171:  
HTTP request sent, awaiting response... 200 OK  
Length: 58238592 (56M) [application/octet-stream]  
Saving to: ‘llm_512.pth.1’  
  
llm_512.pth.1      100%[=====>]   55.54M   55.3MB/s    in 1.0s  
  
2026-01-03 07:49:58 (55.3 MB/s) - ‘llm_512.pth.1’ saved [58238592/58238592]
```

```
/content/drive/MyDrive/mini-vm/mini-vm
/content/drive/MyDrive/mini-vm/mini-vm/trainer
```

```
%cd ..
%cd model/vision_model
!git clone https://www.modelscope.cn/models/openai-mirror/clip-vit-base-patch16

/content/drive/MyDrive/mini-vlm/mini-vlm
/content/drive/MyDrive/mini-vlm/mini-vlm/model/vision_model
fatal: destination path 'clip-vit-base-patch16' already exists and is not an empty directory
```

```
/content/drive/MyDrive/mini-vlm/mini-vlm/model
/content/drive/MyDrive/mini-vlm/mini-vlm
/content/drive/MyDrive/mini-vlm/mini-vlm/trainer
```

```
2026-01-03 07:56:21.372152: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:348]
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1767426981.392359    9064 cuda_dnn.cc:8579] Unable to register cuDNN factory
E0000 00:00:1767426981.398598    9064 cuda_blas.cc:1407] Unable to register cuBLAS factory
W0000 00:00:1767426981.414381    9064 computation_placer.cc:177] computation placer a
W0000 00:00:1767426981.414426    9064 computation_placer.cc:177] computation placer a
W0000 00:00:1767426981.414430    9064 computation_placer.cc:177] computation placer a
W0000 00:00:1767426981.414435    9064 computation_placer.cc:177] computation placer a
所加载VLM Model可训练参数: 3.213 百万
```

```
Epoch: [1/1] (100/37207) loss:3.405802 lr:0.000439992871 epoch_Time:628.0min:  
Epoch: [1/1] (200/37207) loss:3.444293 lr:0.000439971483 epoch_Time:619.0min:  
Epoch: [1/1] (300/37207) loss:3.208273 lr:0.000439935839 epoch_Time:615.0min:
```